BioFauna: Mediterranean taxon identification by retrieval over a frozen BioCLIP-2.5 ViT-H encoder

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Taxonomic sources: published Mediterranean checklists (GROC/OPK; Ballesteros 2007; Salvador et al. 2022) and community curators

Repository: https://github.com/yespi/biofauna
Live system: https://fotofauna.yespi.es
Manuscript version: 2026-10-11 (article aimed at Biodiversity Data Journal / PeerJ; OOS 83.22 %; unified catalogue expected ≈3,109 spp)

> **Naming.** Developed as YOLOFauna (2024–mid-2026); renamed BioFauna when production settled on BioCLIP retrieval rather than YOLO detection. Provenance: `docs/HISTORY.md`.

Abstract

BioFauna identifies Mediterranean (and incidental adjacent) taxa from photographs by **retrieval** over a regional gallery, without training a closed-set classifier. The production encoder is **BioCLIP-2.5 ViT-H/14** (frozen; 1024-d embeddings). Queries are encoded (global frame fused with a 65 % centre crop), searched with **k-NN (k = 15)** and a tempered vote aggregator, optionally re-weighted by a geographic prior, mapped to calibrated probabilities and, when the margin is weak, **abstained** to genus, family, or a curated indistinguishability group.

Current figures (`dataset/stats.json` `updated=2026-10-11T08:00:50`; `/health` 11 Oct): live gallery **1,216,896** embeddings / **4,643** species in the FAISS index; target catalogue **3,084** embedded species (**3,083**), with unification expected at ≈**3,109** spp; field panel **78,145** rows with out-of-sample species accuracy **83.22 %** (genus **87.96 %**, family **91.04 %**; `calibration.field_acc`). AutoID publishes at species threshold **0.83** (precision ≈96 %, coverage ≈66.5 % on the full eval; audited historical precision **96.6 %**, n=493). Baselines on 78,180 rows (2026-10-03): nearest centroid **74.73 %**; plain k-NN **79.86 %**. A systematic search for gains from training on this embedding space (LoRA, QLoRA, ArcFace, etc.) **did not beat** frozen retrieval. We publish the experiment ledger, taxon IDs, prototype centroids and a self-hostable identifier; photographs and per-photo embeddings are **not** redistributed.

Keywords: BioCLIP-2.5, ViT-H, k-NN, fine-grained visual classification, marine biodiversity, citizen science, taxonomic abstention, calibration, Mediterranean Sea

1. Introduction

The Mediterranean holds a disproportionate share of marine biodiversity in a small basin, while taxonomic capacity is scarce. Citizen-science networks (Minka, iNaturalist) collect photographs faster than experts can identify them. Global classifiers help, but regional endemics, cryptic invertebrates, and language-specific field knowledge remain a bottleneck.

BioFauna is the identifier behind FotoFauna. Design choice: **do not train a closed-set CNN**. Use a biology-pretrained vision encoder and a **regional reference gallery** that can grow species by species without retraining.

Contributions

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1. A production retrieval stack (frozen ViT-H + k-NN + calibration + hierarchical abstention) running on consumer GPU and publishing high-confidence IDs to Minka.
2. An **observation-stratified** evaluation protocol, including a gallery-similarity leak check after a 42.7% self-duplicate incident.
3. A **ledger of experiments** (this paper §4 and `docs/EXPERIMENTS.md`): each trial has a hypothesis, a result, and whether it was kept.
4. Open reconstruction artefacts: taxon IDs, prototype centroids (4,543 × 1024), calibrators, geo priors, cryptic-pair lists — not the photo corpus.

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2. System

2.1 Encoder

Property	Value
Checkpoint	`hf-hub:imageomics/bioclip-2.5-vith14`
Architecture	ViT-H/14, ~632M params, 1024-d L2-normalized embeddings
Input	224×224
Training in BioFauna	**Frozen**

2.2 Identification (production, 2026-09-23)

1. Embed the query; fuse **global frame + 65% centre crop** (ROI fusion; replaces an earlier 90% TTA).
2. Retrieve **k=15** gallery neighbours (cosine / inner product via FAISS when the full gallery is present; nearest-centroid over published prototypes otherwise).
3. Aggregate neighbour scores with **tempered votes** Σ `exp(max(s,0)/T)`, **T=0.05**, plus a prototype-similarity boost. Each species contributes **at most 3** of the 15 neighbours to the vote (`KNN_CLASS_CAP=3`, K23), so a species with a huge gallery cannot outvote a sparse one by sheer count.
4. Optional **multiplicative geographic prior** when GPS is present.
5. **Abstain** when the top-1/top-2 margin is small and taxa share genus/family, or when the pair/genus/group is in `dataset/taxonomic_exceptions.json`.
6. Map k-NN features → **P(correct)** with logistic regression (`dataset/calibration.json`).

AutoID on FotoFauna publishes when calibrated **p ≥ 0.80**. With the calibrator refit on leak-free data (`created=2026-09-23T08:22`, fit 8,509 / test 3,864 rows, species-disjoint split) that threshold gives **97.0% precision at 80.6% coverage** on the test split (closed-set; the recent-observation KPI is lower, O16). The August estimate (95.3% at 57.4%) is superseded.

This public repository ships **prototype centroids** (`data/patterns/<slug>/prototype.npy`). That is enough for a nearest-centroid demo. Full k-NN accuracy requires rebuilding per-photo embeddings locally (see `docs/dataset.md`).

2.3 Data (what exists vs what we release)

Layer	Production	This repo
Photographs	~1.06M on disk (Minka, iNaturalist, GBIF, …)	**Not released** (licence)
Per-photo embeddings	1,118,353 vectors	**Not released** (size + derived from photos)
Prototype per species	4,543 × 1024 float32	**Released** (~15 MB)
Taxon IDs / names	2,985 catalog rows	**Released** (`dataset/catalog.json`)
Calibrator, geo priors, cryptic pairs, exceptions	live JSON	**Released**

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Names are cross-checked with **WoRMS** (accepted name stored beside the stable slug). Images are filtered for format/resolution; field-guide plate crops that mixed several taxa under one label were quarantined (1,088 photos / 107 species in the August 2026 audit).

Tiers (catalog, not gallery size):

Tier	Meaning	n taxa
0	Heterobranchs (expert collaborators' core group)	634
1	Other marine	1,527
2	Terrestrial / incidental in marine photos	824

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3. Evaluation protocol

Headline numbers use held-out **observations** (an encounter, often several photos of one organism), not a random photo split. Photo-level 80/20 inflates k-NN and metric-learning numbers by ~10 pp because citizen-science bursts are near-duplicates.

Rule	Why
Stratify by observation ID	No burst leakage
Reject a candidate if it is too similar to the live gallery	Catches missing IDs after path migrations
Same protocol across ablations	Comparable McNemar
Do not mix harvests when citing a delta	Different species mix ≠ model gain

Incident (2026-09-23). The August gate below was itself ineffective: it compared a TTA-averaged (orig + 90% crop) query embedding with gallery embeddings at cosine >0.999; an identical photo scores ~0.99 that way. 50.8% of the evaluation set had re-entered the gallery. The gate now compares the **global** embedding (as stored in the gallery) at **≥0.98**, and an audit script re-checks every evaluation row after any gallery growth (O18).

Incident (2026-08-25/26). An observation-ID denylist pointed at an abandoned path and silently matched nothing. **42.7%** of a 22,332-photo harvest were already in the gallery. At k=15 the headline only moved ~1 pp; at small k the curve was absurd (accuracy rising toward k=1). Fix: embed each candidate and drop near-duplicates. Kept as a standing harvest gate.

Two labelled metrics in this paper

Label	Cohort	Species top-1	Use
A — TTA-era freeze	`harvest_calib` n=12,788 (Aug 2026, leak-checked)	75.97% (later 77.77% with inference stack; 79.10% jsonl after densification)	Comparable ablations in §4.1–4.2
B — live calibrator	`calib_raw_t05` n=19,087 (2026-09-14)	**85.78%**	Current production report

| **C — leak-free** | `calib_raw_t05` n=12,373 (2026-09-23, audited, ≥0.98 copies removed) | **88.11%** | Current production report (§5) |

B is larger, covers more hard/rare species, and uses T=0.05 + later gallery quality work. **A is not "worse engineering"; B is not a 10-point magic leap on the same photos.**

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4. Experiments

Each row: **hypothesis → result → contribution to the project.** Details and McNemar counts: docs/EXPERIMENTS.md.

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4.1 Kept (in production)

ID	What we tried	Result	What it contributed
K1	Replace ViT-L gallery with **frozen ViT-H**	+6.8 pp species (63.9% → 70.6% on the then cohort)	Largest single modelling gain. Production encoder.
K2	k-NN grid, observation-stratified	**k=15** beats k=8 and k≥20	Production k.
K3	Unify SSD + HDD-archive photos and re-embed	71.7% → 75.8% on leak-checked n=12,788	Closed the archive gap (see `docs/archive/ARCHIVE_GAP.md`). Completeness > adapters.
K4	Logistic calibration on 10 k-NN features	ECE ~0.03 vs ~0.09 raw cosine	Interpretable AutoID threshold.
K5	Hierarchical abstention (margin + expert pairs/genera)	Genus/family fallbacks ~89% when used; species top-1 slightly conservative	Safer citizen-science output.
K6	90% centre-crop TTA, then **65% ROI fusion**	TTA +0.21–0.75 pp; ROI fusion +1.63 pp vs global-only on n=12,788	Production query embedding.
K7	Late fusion of multiple photos in one observation	+0.79 pp full set; +3.15 pp on the 25% multi-photo subset	Production when N>1.
K8	Fisher / PCA+LDA **local subspace** on documented cryptic pairs	+0.14 to +0.36 pp on the official harvest	Scoped re-rank; not a new encoder.
K9	Inter-genus same-family local subspace	Official re-harvest **77.77%** (cohort A + inference stack)	Production second trigger.
K10	Gallery densification (more/better reference photos, frozen encoder)	Cohort A jsonl **79.10%**; later quality swaps (curators, Q≥8, seagrass)	Data, not weights. Live gallery 848,883.
K11	Tempered k-NN aggregator T=0.05	McNemar n=8,000: **+7.81 pp**, 671/46	Production scorer (cumulative with K10).
K12	MiniCPM gate on two confusion pairs (Branchiomma/Myxicola, Cerianthus/Pachycerianthus)	Small-n McNemar 12/1, p=0.003	Gated sidecar, not a general VLM re-ranker.
K13	Nomenclature audit (WoRMS + GBIF)	85 accepted-name fields; 4 duplicate-slug merges	Catalog hygiene; slugs never used as WoRMS names.
K14	Expert indistinguishability list + sponge group abstain	Extra pairs from eval confusion (≥33% reciprocal)	Decision layer where vision saturates.
K19	Grow reference photos to ~1,000/species where the sources have them + full per-species re-embed	McNemar n=18,000: **+0.57 pp** (313/211, p=1e-5). By species: grew ≥+300 vectors **+7.7 pp**, +100–299 **+12.0 pp**, +1–99 +3.4 pp; species that did not grow −0.63 pp (dilution)	Data works where it exists; 77% of remaining errors sit in species that did not change.
K20	Second growth wave (121 species, 21.9k photos), rebuild to 1,118,353 vectors	McNemar n=21,000: **+0.22 pp** (84/38, p=5e-5); touched species **83.3%→89.5%** (83/1); 31 improve, 0 worsen	In production 2026-09-21.
K21	Independent second opinion for the closed-set retriever: BioCLIP zero-shot over a regional checklist	300 Mediterranean bird observations: BF≥0.85 **and** zero-shot≥0.80 agree → **98.5% precision at 69% coverage** (BF alone 91.2% at 79%). All groups: 95.9% at 49% with a rescue tier vs BF alone 93.8% at 48%	Open-set guard for AutoID (birds in production; other groups pending).
K23	Per-species vote cap in the k-NN aggregator (`KNN_CLASS_CAP=3`)	McNemar n=12,000 held-out obs: **+1.13 pp** (92.33%→93.47%; 228 fixes / 92 breaks); **177 species improve / 70 worsen**; gains largest for species with <25 gallery photos (+3.2–3.5 pp). cap1 −0.19, cap2 +0.82, cap4 +1.08, cap5 +0.94 pp	In production 2026-09-22 07:17 CEST. (Measured before the O18 purge; the relative gain is what counts.)
K22	Promotion criterion counted in species	173 species improve / 153 worsen / 2,186 unchanged (previous→current index)	Report the per-species balance next to Δ and p.

4.2 Rejected (do not repeat as-is)

ID	What we tried	Result	What it contributed
R1	QLoRA ViT-L, trainable projection	1.7% species	Compatibility/fragility lesson; not a BioCLIP failure in general.
R2	QLoRA BioCLIP-2 ViT-L vs ViT-H prod	Architecture mismatch (768 vs 1024-d)	Discarded before eval.
R3	Triplet on ViT-H (8 variants)	−0.7 to −7 pp	Metric learning on this gallery memorizes.
R4	ArcFace on frozen ViT-H	Tie with k-NN (~71.4 vs 71.6)	No reason to replace retrieval.

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ID	What we tried	Result	What it contributed
R5	LoRA+ArcFace 100 spp	+0.0 pp OOS	Internal splits lied.
R6	LoRA last-4-blocks + ArcFace, 1,358 spp	**−31.2 pp** on full eval	Fine-tuning a subset poisons the shared space.
R7	Frozen-backbone linear head	−0.6 to −1.1 pp OOS (train mini-set had shown +2.6)	Train/eval disagreement; no cutover.
R8	Weight expert-guide crops in the gallery	−0.9 pp	OCR plates ≠ underwater photos.
R9	Near-duplicate burst removal (cos>0.99)	−1.6 pp	Bursts help k-NN.
R10	Outlier filtering on embeddings	−0.21 to −1.45 pp	Removed intra-species variation.
R11	Widen same-genus abstention margin	9:1 cost of true species IDs vs recovered errors	Production τ stays tight.
R12	“Prefer epibiont” re-rank	−0.23 pp (64 fixed / 116 broken)	Host photos look like the partner in top-k.
R13	Scoped SupCon re-ranker on 20 cryptic pairs	Val loss diverged from epoch 1; kill-switch **before** the eval set	Third independent architecture, same failure mode.
R14	Horizontal-flip TTA	−0.13 pp	Asymmetry is signal.
R15	Per-species prototype weight by dispersion	−0.09 pp	Global dispersion ≠ this query.
R16	Family-consensus penalty on k-NN	−0.41 pp, p=0.0005	Majority vote is already wrong on convergent morphologies.
R17	Attention-guided crop (Bucket B)	+0.04 pp, p=0.77	Noise.
R18	Phylum/class consensus filter	All thresholds net-negative	80% of cross-group errors are already in k-NN.
R19	Ancestral subspace for spp with <5 refs	1 addressable eval observation	Closed before GPU spend.
R20	Background/substrate neutralization	−0.92 pp vs own control; −3.48 vs full stack	Habitat is signal for epibionts.
R21	Extra geo prior from Minka point clouds on top of production geo	Not significant; 4/5 OOF folds chose strength 0	Confusable species co-occur.
R22	DINOv2 fusion (scripts misnamed “DINOv3”)	−6.75 pp vs production	Prototypes-only DINOv2 is far weaker here.
R23	Morphological multi-prototypes (k-means)	Not significant; 4/5 folds blend=0	Single centroid already sufficient for the boost term.
R24	Selective marine re-embed (36 spp)	~0 pp	No cutover.
R25	Seagrass VLM / generic densification	Null	Quality swap helped *Posidonia*, not *Cymodocea*.
R26	MiniCPM “pure subject” filter on fauna (16 spp)	97.5% already “pure”	Does not transfer from seagrass.
R27	Broad Q≥8 photo swap on already-exhausted taxa	Very low yield on some batches; high yield on others	Quality helps **when** better photos exist; not a universal lever.
R28	Raise the confidence threshold to cut confident errors	Live backtest n=184: precision flat at 90–92% for p≥0.80…0.95; only coverage falls (52%→11%)	Calibration saturates; not a lever.
R29	Require iNaturalist CV agreement before publishing	Of 81 BF answers ≥0.85: iNat agreed on 51 (BF right 96%), returned nothing on 29 (BF right 83%), disagreed on 1; ~4 correct answers blocked per error avoided	Replaced by domain guards + zero-shot consensus (1-day trial, audit pending).

4.3 Operational failures (not model ideas, still public)

These are included because they change how numbers should be read.

ID	Failure	Effect	Contribution
O1	Calibration harvest leaked into the gallery (Aug 2026)	Inflated low-k curves	Similarity dedup is mandatory.
O2	FAISS index vs `species_ids` desync after `/reload` (1 Sep 2026)	Live IDs assigned terrestrial queries marine names at 85–100% confidence; **disk evals valid**	Reload index and labels together.

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ID	Failure	Effect	Contribution
O3	Admin metrics reading frozen snapshot files	Panel showed stale/in-sample accuracy	Live metrics from the current jsonl.
O4	Empty `--species` list after nested SSH (13 Sep 2026)	Quality-swap ran on extra taxa for ~35 min	Pass slug files, never interpolated lists.
O12	The serving process translated FAISS labels with the live species folder list, not the index's own name list; three new folders shifted every label while `/health` still reported aligned	Wrong species at 0.92 similarity for hours (2026-09-20/21); AutoID published nothing	Guard cron compares index names ↔ loaded species ↔ folders (extends O2).
O13	An unattended job rebuilt the production FAISS directory and refit calibration before validation finished	Production index overwritten by an unvalidated build	Unattended jobs never write production paths.
O14	The Minka web host returns 403 to the default `python-httpx` User-Agent (login and photo downloads)	AutoID zero publications for days (second recurrence of the missing-header class)	One shared HTTP client; alarm on zero-publication days.
O15	Two agent sessions ran the same cycle in parallel; one promotion overwrote the other; GPU OOM with sidecar + serving + re-embed	Duplicated compute; index/calibration mismatch for ~30 min	Single active session with a written registry.
O16	Closed-set, observer-correlated evaluation: panel 92.4% vs ~80% on 300 recent research-grade observations; 12% of bird observations are species outside the catalog	Panel overstates real-world accuracy	Recent-observation sample as operating KPI.
O18	Evaluation leak gate compared a TTA-averaged embedding with a 0.999 threshold; identical photos score ~0.99, so the gate never fired. Later ad-hoc searches (Wikimedia/GBIF/iNat any grade) had no gate at all	50.8% of evaluation rows were gallery copies; panel 92.4–93.4% vs 88.1% clean; rare-species accuracy on unseen photos 24% hidden as 57%	Global-embedding gate ≥0.98 in every harvester; `leak_audit_calib.py` after every gallery growth; leaked rows kept aside, calibrator refit (2026-09-23).
O17	Dashboard accuracy per species changes only when the eval is re-scored against the serving index	Numbers frozen after promotions	Daily automated re-score; re-score after every promotion.

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5. Current production report (cohort C, leak-free, 2026-09-23)

Level	Accuracy	n
Species	**88.11%**	12,373
Genus	**90.55%**	12,373
Family	**92.46%**	12,373
Tier 0 (heterobranchs)	85.12%	2,446
Tier 1 (other marine)	87.72%	5,633
Tier 2 (terrestrial/incidental)	90.34%	4,294

Per species (2,091 with clean evaluation): 1,429 at 100%, 431 below 80%. 894 catalog species lost their only evaluation rows in the purge and are being re-harvested with the fixed gate. Cohort B (85.78%, n=19,087, 14 Sep) is kept for history; it was affected by the same leak.

Remaining error is dominated by **visual cripsis and morphological convergence** (sponges, filamentous algae, some fishes, seagrass blades), not by “we need LoRA”. Coverage: most catalog species now have ≥1 held-out observation; a small tail is exhausted on Minka+iNaturalist (dozens of taxa with essentially no extra research-grade photos).

Seagrass (n=60 each, 14 Sep): *Posidonia oceanica* 73.3%; *Cymodocea nodosa* 70.0% (quality swap **not** shipped — confusion with *Nanozostera*); *Nanozostera noltii* 86.7%; *Zostera marina* 75.0%. The pair *Z. marina* ↔ *N. noltii* was added to expert abstention after 20/120 cross-confusions.

5.1 Field report by block (2026-10-02, all figures measured)

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Gallery. 1,118,321 vectors / 4,543 species from iNaturalist 733,789 photos, Minka 372,347, GBIF 39,570, DORIS/FFESSM 5,091, Wikimedia 4,197, WoRMS 1,319, SeaSlugForum 817, FishBase 646. Vectors per species: 775 species have fewer than 10, 1,058 have 10–29, 450 have 30–99, 1,123 have 100–299, 935 have 300–999 and 202 have 1,000 or more (median 97, maximum 3,223); the gallery is strongly long-tailed.

Field accuracy (species).

Block	Rows	Accuracy	Caveat
Original evaluation set (comparable with all earlier versions)	60,743	**81.18%**	observer-correlated, closed-set
Mining round 1 (after purge)	11,154	85.49%	species with larger pools
Mining round 2 (after purge)	4,688	89.87%	idem
All purged	**76,585**	**82.34%**	rise vs. 81.18% is mostly composition
Panel, by realm (78,180 rows, no observer purge): marine	65,813	80.54%	realm is metadata, it does not filter publication
terrestrial	9,853	93.30%	
birds	2,027	92.25%	
non-native (Lessepsian etc.)	487	82.34%	small n

On the purged 76,585 rows the realm figures are marine 80.37%, terrestrial 93.37%, birds 92.22%, non-native 82.08%.

AutoID in practice. On 493 historical publications audited on 2026-09-29 the precision was **96.6%**, by calibrated-confidence band: 0.83–0.85 → 85.2% (n=27), 0.85–0.90 → 93.2% (n=59), 0.90–0.95 → 95.5% (n=155), ≥ 0.95 → 99.2% (n=252). A replay of 726 publications gave 94.6% top-1 and 97.96% precision at p ≥ 0.83 (n=589). Photos AutoID does not publish are not errors: in the last hour measured, of 1,429 candidate observations 1,373 stayed below threshold, 50 were skipped because the same observer already had that taxon in an album that day, and 81 failed the photo-quality gate.

Where the errors are. Worst species in the panel (all with ≥ 11 evaluation photos): *Treptacantha nodicaulis* 0/20 (confused with *Gongolaria barbata*), *Turbonilla pusilla* 0/18 (with *T. lactea*), *Pegusa nasuta* 0/18 (with *Solea solea*), *Forskalia tholoides* 0/13 (with *F. edwardsii*), *Arion rufus* 0/29 (with *A. ater*), *Phyllidiella granulata* 0/16 (with *Phyllidiopsis krempfi*), *Carduelis carduelis* 35% of 40 (with *Spinus spinus*). Most frequent confusions (errors): *Petalifera petalifera → P. ramosa* 31, *Patella aspera → P. ulyssiponensis* 27, *Tamarix africana → T. gallica* 26, *Elysia marginata → E. ornata* 24, *Pyracantha coccinea → P. crenulata* 24, *Dictyota implexa → D. dichotoma* 23, *Ulva rigida → U. lactuca* 22, *Diplodus sargus → D. cadenati* 21. Most are congeneric pairs with very similar morphology; this report does not test why each pair fails.

What these numbers do not say. They are closed-set and observer-correlated; the operating KPI is the later third-party reaction to published identifications, which for crop rescues does not exist yet. Per-species accuracy was not broken down by gallery size in this report.

Useful accuracy under hierarchical abstention (2026-10-03, offline, calibrated probabilities). On the same 78,180 field rows, using the calibrated probabilities of the three levels (species, genus, family; calibration fitted on these rows, hence mildly optimistic), a cascade "publish species if p ≥ 0.83, otherwise genus if p_genus ≥ τ, otherwise family if p_family ≥ τ" gives:

Policy	Photos with a publishable ID	Precision of that set
Species only, p ≥ 0.83	65.1%	95.9%
+ genus, τ = 0.83	77.2% (+12.1 pp)	95.6% (added rows: 93.7%)
+ genus, τ = 0.90	74.1% (+9.0 pp)	95.9%
+ genus 0.83 + family 0.83	83.8% (+18.7 pp)	95.3% (added family rows: 91.7%)
+ genus 0.90 + family 0.90	79.9% (+14.8 pp)	95.9%

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Family-level publication only pays off at τ ≥ 0.90; at 0.83 its precision drops to 91.7%. Species-level top-1 accuracy (82.46% panel, 77.5% Tier 1) is unchanged by construction: the gain is in photos that receive a correct identification at the coarsest rank the model can defend. Live crop-rescue and genus-level AutoID publications are still too few and too rarely reviewed by third parties to validate these figures. **Out-of-sample check (2026-10-03):** on 575 recent research-grade Minka observations never used for calibration (leaks with top-1 similarity ≥ 0.98 excluded), species-level p ≥ 0.83 covered 73.2% of photos at 91.7% precision (4 pp below the in-sample 95.9%), and the genus step added +7.3 pp of photos at 85.7% precision (p_genus ≥ 0.83), +4.3 pp at 88.0% (≥ 0.90) and +2.4 pp at 92.9% (≥ 0.95). The in-sample table above is therefore optimistic; only the p_genus ≥ 0.95 rule reaches species-level precision out of sample. A second, independent sample (692 further observations) gave the same picture; pooled (n = 1,267): species p ≥ 0.83 covers 71.1% at 91.2% precision, and the genus step adds +7.4 pp at 88.3% (≥ 0.83), +4.4 pp at 87.5% (≥ 0.90) and +2.4 pp at 90.3% (≥ 0.95; n = 31 added photos), so the gain from hierarchical abstention is real but small (≈ +2 to +7 pp of photos) and slightly less precise than species-level publication.

5.2 Related work and baselines

Position. BioFauna sits between two lines of work. Vision-language models pre-trained on the tree of life (BioCLIP [1], built on CLIP [4]) give embeddings that separate species without task-specific training; large citizen-science corpora (iNaturalist [5]) provide the labelled photographs. Many deployed systems fine-tune a classifier on such data; BioFauna instead keeps the encoder frozen and does **k-NN retrieval over a regional gallery** with calibrated abstention, so that adding or fixing a species is a gallery edit, not a retraining run. Metric-learning losses (triplet [13], FaceNet [12], supervised contrastive [6]) and parameter-efficient fine-tuning (LoRA [3], QLoRA [2]) were the obvious alternatives; §4.2 reports that none of them beat frozen retrieval on this gallery.

Baselines in this report. (a) *Nearest centroid* over the published prototypes (public demo): **74.73%** top-1 on the 78,180 field rows (2026-10-03). (b) *Plain k-NN with majority vote* (k = 15, exact search on the production index), without the tempered aggregator, class cap, geographic prior or calibration: **79.86%** (plain 1-NN: 79.10%), against 82.46% for the full production system on the same rows, i.e. the decision layer adds about 2.6 pp over plain voting and the retrieval itself about 5.1 pp over a prototype-only classifier. (c) *Fine-tuned or metric-learning heads*: scored in §4.2 on earlier evaluation sets, rejected. (d) *The iNaturalist computer-vision suggestion*, as a baseline for AutoID: measured **partially** on 2026-10-03 (n = 98 field photos; the run stopped at an HTTP 429 rate limit before the planned 300 and was not retried): iNaturalist CV gives **55.1%** species and **67.3%** genus top-1, against **84.7%** for BioFauna on the same photos (BioFauna right and iNaturalist wrong: 31; the reverse: 2). With n = 98 the interval is wide (about ±8 pp), so this is indicative, not a headline figure; completion to n = 300 is pending.

Baseline	Evaluation (rows)	Species accuracy	Status
Full BioFauna (production, 5-Oct panel)	78,180	82.46 %	measured
Nearest centroid (published prototypes)	78,180 (2026-10-03)	**74.73 %**	measured
Plain k-NN, majority vote (k = 15)	same rows	**79.86 %** (1-NN: 79.10 %)	measured
Current BioFauna (panel after invasoras 10 Oct)	78,145	**83.22 %**	measured (`stats.json` 0.8322)
iNaturalist CV suggestion	98 (partial; target 300)	55.1 % (BF 84.7 % on same)	partial

The decision layer adds ≈2.6 pp over plain k-NN voting and retrieval ≈5.1 pp over prototypes alone (measured 2026-10-03 on 78,180 rows). The iNat baseline remains incomplete (n = 98 after HTTP 429).

5.3 Clean evaluation, index promotions and methods not adopted

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Evaluation set. The field panel (78,145 rows, species accuracy 82.463 % on 2026-10-05) contained duplicated photos (12,416 rows), multi-subject photos (2,933) and rows whose own photo was in the gallery. The *clean v2* evaluation (62,796 rows) gives 82.841 % (Tier 1 77.93 %). A full leak audit leaves **54,878 leak-free rows (live index 80.63 %, Tier 1 76.42 %)**; every candidate below is also reported on the leak-free rows (thresholds 0.95 and 0.995).

Promotions to production (each with backup and rollback script; McNemar on paired rows, per-species guard: no species with n≥10 loses ≥3 photos or >30 points).

Date	Change	Panel (78,145)	Leak-free	Fixes / breaks	Guard
2026-10-05 18:45	*Clean combo*: 58-species re-embed batch + block E + small harvest, minus 4 guard species and the vectors responsible for 3 "thief" species	82.463 → 82.626 % (+0.163 pp, p=2e-11)	+0.261 pp (≥0.95) / +0.212 pp (≥0.995)	240 / 113 (panel); 212 / 69 and 213 / 81 (leak-free)	clean
2026-10-06 02:50	*Harvest batches 01+02* (71 species, added iNat photos, thieves removed iteratively)	82.626 → 82.85 % (+0.223 pp, p=7e-27)	+0.503 pp (≥0.95) / +0.424 pp (≥0.995)	217 / 43	clean

Index after both: 4,543 species, 1,132,767 vectors, aligned. Harvest batches measured against the new base give diminishing returns (batch 03: +0.088 pp; batch 04: +0.024 pp): each species contributes only 8–44 new accepted photos because the limit is the number of unseen research-grade observations, not the filters. **Sibling effect**: adding photos to a species lowers accuracy of its close relatives (e.g. *Pyracantha coccinea → P. crenulata* 32→19 of 40), so batches are merged and the "thief" species removed iteratively until the guard is clean.

Finding on gallery coverage. At 2026-10-06 the disks hold 1,223,506 photos (SSD + archive) but the index has 1,132,767 vectors: **96,392 photos in 867 species were never embedded** (photos harvested after each species' last embedding; e.g. *Papaver rhoeas* 33 vectors vs 1,000 photos). Full per-species re-embedding of the 715 species still below 100 % (72,298 photos, 3,845 panel errors) is in progress, with the same leak filters (embedding similarity ≥0.98 to gallery, ≥0.95 to any evaluation photo and the whole observation, centroid and own-prototype checks).

Accuracy by gallery size (clean panel, 62,248 rows): 0–25 photos 68.6 %; 25–50 76.0 %; 50–100 65.7 %; 100–200 82.0 %; 200–500 82.2 %; 500–1,000 86.1 %; ≥1,000 89.2 %.

Negative or null results (not adopted).

Idea	Result
Removing discordant prototype vectors (1,188)	−0.023 pp (25 fixes / 43 breaks, p=0.039 against; 32 species worse)
DINOv3 as tie-break on 16 confusable pairs	58.9 % vs 59.6 % for BioFauna; tie-break null at every δ
Vision-language comparator (MiniCPM-V)	51.9 % vs 58.3 % for BioFauna (108/579 valid answers)
Grounding DINO and fine-tuned YOLO26n for the attention crop	Worse than the current fusion (GDINO 74.0 vs 80.85 % on 4,000 photos; YOLO + best = identical, 14/14)
Under-exposure enhancement on photos the whole-image pass does not publish	17 of 392 would publish; 15 correct, 2 wrong (88.2 %); mostly already covered by the attention rescue
Colour (Lab histograms, k-means palette) as tie-break	Net negative at every δ; on published photos it fixes 0 and breaks 4–13
Zone prior (local species counts) as tie-break	Plain argmax loses; with a ≥10× condition +3/−0 on the third-party sample (too small to conclude)
Posidonia/seagrass re-embed; local small-gallery harvest	+0.005 pp (3/0); +0.008 pp

Automatic identification (AutoID) changes. The "avoid sibling observations of the same user and day" rule was removed on 2026-10-04: on 349 published observations, the 42 (12 %) substituted by an alternative were correct in 3 of 11 judged cases (27 %) versus 102 of 105 (97 %) for the non-substituted ones; an alternative's raw similarity had also been compared with the minimum as if it were a calibrated probability (the cause of a real misidentification). A 14-day replay (500 publications) of the current cascade (whole photo; else blue-cast correction; else attention crop with a strict guard) shows 97 % accuracy (77/79 judged) where it publishes the same species, and corrects 4 of 7 judged changes while breaking 2. The species-level threshold remains 0.83; genus-level publication p≥0.95 is on; the crop-mix publication is paused.

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Limits of these numbers. Judged observations are those with an identification from another user; replays use today's model, not the one at publication time; the harvest gains are on a panel built from the same sources as the gallery, so the leak-free rows are the more reliable estimate.

5.4 Current figures after the invasoras promote (OOS 83.22 %)

Current panel (`dataset/stats.json`, `updated=2026-10-11T08:00:50`; health OK on `:8090`):

Quantity	Value	Source
Eval rows	**78,145**	`stats.json` → `calibration.n_samples`; `calib_raw_t05.jsonl`
Species OOS accuracy	**83.22 %** (0.8322)	`stats.json` `overall_accuracy` / `calibration.field_acc.species`
Genus	**87.96 %** (0.8796)	`stats.json` `calibration.field_acc.genus`
Family	**91.04 %** (0.9104)	`stats.json` `calibration.field_acc.family`
Live index (health)	**1,216,896** vectors / **4,643** spp	`stats.json` `faiss_vectors` / `live_health.species` (invasoras promote 10 Oct)

Panel series (78,145 rows):

Milestone	Species OOS	Source
Pre–clean combo (5 Oct)	82.463 %	§5.3
After *clean combo*	82.626 % (+0.163 pp)	§5.3
After harvest batches 01+02	82.85 % (+0.223 pp)	§5.3
After *all08it2* (8 Oct 23:42)	**83.14 %** (0.8314)	`agent-bus/inbox/claude/20261008-234238-robotin.md`
After invasoras *todoselmejB* (10 Oct 09:51) / *invfull2* (11:51)	**83.22 %** (0.8322)	`logs/cola_gpu_4b_20261010.log.promote_*`; `stats.json`

The *all08it2* promote measured **+0.294 pp** (fix 353 / break 123; `artifacts/decide_promote_calib_raw_t05_stg_all08it2_20261005.json`). The 10 Oct invasoras promote used the **invasoras criterion** (delta +0.083 pp on *todoselmejB*; backup `dataset/bak_promote_invasoras_20261010_090651`).

AutoID (species threshold 0.83). On the full eval, with the live logistic calibrator (`calibration.json` coef/mu/sd on k-NN features in `calib_raw_t05.jsonl`):

Threshold	Coverage	Precision	n
p ≥ 0.83 (full eval)	**66.42 %** (≈ 66.5 %)	**95.97 %** (≈ 96 %)	51,906 / 78,145
p ≥ 0.80 (operating_points, test split)	69.9 %	95.4 %	16,409
p ≥ 0.85 (operating_points, test split)	65.3 %	96.2 %	15,326

The calibrator's `operating_points` table does **not** include exactly 0.83 (it jumps from 0.80 to 0.85); the full-eval 0.83 point is what production uses. Audited historical AutoID publications: **96.6 %** precision (n=493; `stats.json` `kpi_referencia.autoid_real`).

Discarded methods (measured; not adopted). Complements §4.2 and §5.3:

Method	Result	Source
SigLIP (2nd backbone, top-5 tie-break)	27 fixes / 36 breaks, n=12,298, p=0.31 (net negative)	`artifacts/exp_altbackbone_siglip_20260930.json`
DINOv2 fusion (scripts misnamed DINOv3)	**−6.75 pp** vs production	§4.2 R22

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Method	Result	Source
DINOv3 as tie-break (16 pairs)	58.9 % vs BioFauna 59.6 %; null tie-break at every δ	`artifacts/exp_dinov3_pares_20261005.json`; §5.3
VLM (MiniCPM-V)	51.9 % vs BF 58.3 % (108/579 valid answers)	`artifacts/exp_vlm_compara_pares_20261005_minicpm-v_latest.json`; §5.3
BioCLIP-2 ViT-L (2nd backbone)	+0.40 pp on touched rows (+0.064 pp on the eval), p=0.06	`artifacts/exp_altbackbone_bioclip2_20260930.json`; `experimentos/EXPERIMENTOS_HACIA_100_20260930.md` E9
YOLO crop (YOLO26n vs current fusion)	74.97 % vs 80.85 % (n=4,000)	`artifacts/exp_det_ab_20261005_yolo.json`; §5.3
ArcFace / LoRA	R4 ties k-NN; R5 +0.0 pp OOS (100 spp); R6 **−31.2 pp** (1,358 spp)	§4.2 R4–R6
Embedding-outlier pruning	−0.21 to −1.45 pp	§4.2 R10
Discordant-prototype cleaning	−0.023 pp	§5.3

Generated figures (matplotlib, 2026-10-09): `../figures/fig_serie_oos_promotes.png`, `fig_acc_especie_genero_familia.png`, `fig_autoid_precision_cobertura.png`, `fig_metodos_descartados.png` (sources in `../figures/fuentes_cifras_20261009.json`).

5.5 iNaturalist confirmation queue (`inat_review_decisions`)

To close the AutoID → community loop, FotoFauna logs review decisions on iNaturalist observations in **`inat_review_decisions`** (shared DB `fauna`, container `postgres-global`). As of **10 Oct 2026** there are **72** rows (read-only; this report does not write to production).

Metric	Value	Source
Total rows	**72**	`SELECT COUNT(*) FROM inat_review_decisions`
`action=confirm`	**27**	same table
`action=discard`	**45**	same table

Manuscript placeholder: when the queue exceeds ~200 verified final taxa, we will publish a summary table (BF proposal vs agreed taxon, latency, observers) for PeerJ/BDJ. Until then, cite only the counts above; do not infer queue precision.

5.5 Update 2026-10-15: accuracy by gallery size, and a zone/month prior

Production index p15. Production now serves the **p15** index (promoted 2026-10-14 17:04 CEST): **2,988 species, 1,289,637 vectors** (previous p13: 3,113 species, 1,300,986 vectors; the drop in species comes from the barrier that keeps in the index only catalogue species with a leak-free evaluation set). Leak-free accuracy: **81.0 %** species top-1 on **44,529** photos from observations without leakage (observer- and observation-disjoint frozen test, `overall_accuracy_clean`, 2026-10-12). On the p15 promote the leak guard measured a net change of −0.18 pp on the leak-free rows (81.73 → 81.55 %, a different run/base from the 81.0 % headline), so the promote bought catalogue hygiene, not accuracy. New in §5.5: accuracy by gallery-size quartile and the case for a zone/month prior.

Accuracy by gallery-size quartile (measured, CPU, read-only). We joined the leak-free keys (`eval_limpio_v3_keys_20261012.json`, 44,529 keys) with the per-photo records of `calib_raw_t05.jsonl` and the per-species vector counts of the p15 index (`species_ids.npy`, 1,289,637 vectors, 2,988 species). 44,480 photos match; 58 of them belong to species without vectors in p15 and are excluded from the table, leaving 44,422. Photos are ordered by the number of gallery vectors of their true species and cut into four equal groups:

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Quartile	Vectors of the true species	Photos	Species top-1
Q1 (smallest galleries)	5–167	11,105	**73.97 %**
Q2	167–399	11,106	81.59 %
Q3	399–816	11,105	86.86 %
Q4 (largest galleries)	816–3,208	11,106	86.79 %

Overall, this scoring file gives 82.20 % on the 44,480 matched photos (it holds the post-p15 decisions; the 81.0 % headline is the `stats.json` frozen-test figure, so the two should not be mixed). Reading: accuracy rises with gallery size up to about 400–800 vectors and then plateaus; the smallest quartile loses about 13 pp against the largest. This is correlational (rare species are also harder and have fewer, less diverse photos), so it does not prove that adding photos alone would close the gap. It does justify prioritising harvest on species with fewer than ~170 vectors, which also lifts Limitation 10.

Zone/month prior (proposal and what is already measured). Presence-only spatio-temporal priors improve fine-grained image classification when the location and date of the photo are known (Mac Aodha et al., 2019 [14]). BioFauna already applies a multiplicative geographic prior (§2.2). Two internal results bound the expected gain: a purely temporal (seasonal) prior was discarded at full scale (76.84 → 76.64 %), and a zone prior used as tie-break only helped under a ≥10× condition (+3/−0, too few cases). We therefore do **not** claim a gain from a joint zone × month prior; the proposed experiment is a leak-free A/B on the 44,529-photo test with the prior computed from catalogue observations excluding the test observations, reported with a McNemar test. No result is available yet.

6. Limitations

1. Species top-1 is not expert-level on cryptic invertebrates; species with thin galleries or thin evaluation sets are much harder than the panel average.
2. The catalogue and evaluation are **Mediterranean-focused** (plus adjacent coasts in the gallery); there is no systematic study outside that region.
3. Public code is a **reconstruction identifier** (prototypes + optional local k-NN), not a dump of the HanSolo production service.
4. The current field panel scores **83.22 %** species accuracy (78,145 rows; `dataset/stats.json`); some rows may still correlate with the gallery, and the recent-observation operating KPI is usually lower than the closed panel.
5. Curator corrections are not yet a closed automatic training loop.
6. **Closed-set retrieval:** species absent from the catalog get a confident wrong answer; mitigated for birds with a zero-shot guard, not yet for all groups.
7. AutoID precision/coverage at threshold 0.83 (≈96 % / ≈66.5 % on the full eval) is from the live calibrator; real precision of published crop rescues still awaits third-party reactions.
8. The iNaturalist CV baseline is measured only at n = 98 (HTTP 429); completion to n = 300 is pending.
9. Several improvement ideas (SigLIP, DINOv2/v3, VLM, YOLO crop, ArcFace/LoRA, outlier pruning) were measured and **not** adopted (§5.3–5.4); the margin over plain k-NN (79.86 %) and centroids (74.73 %) is real but modest.

7. Reproducibility

```
git clone https://github.com/yespi/biofauna.git && cd biofauna
pip install -r requirements.txt
python -m uvicorn src.identify_service:app --host 0.0.0.0 --port 8090
```

Downloads BioCLIP-2.5 from Hugging Face on first run. Uses `data/patterns/` prototypes. To rebuild the full gallery: download photos with taxon IDs in `dataset/catalog.json`, run `scripts/reembed_vith.py`, place `embeddings.npy` beside each prototype.

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Licence: MIT for code and released JSON/prototypes. Photograph copyright remains with original observers and platforms.

---

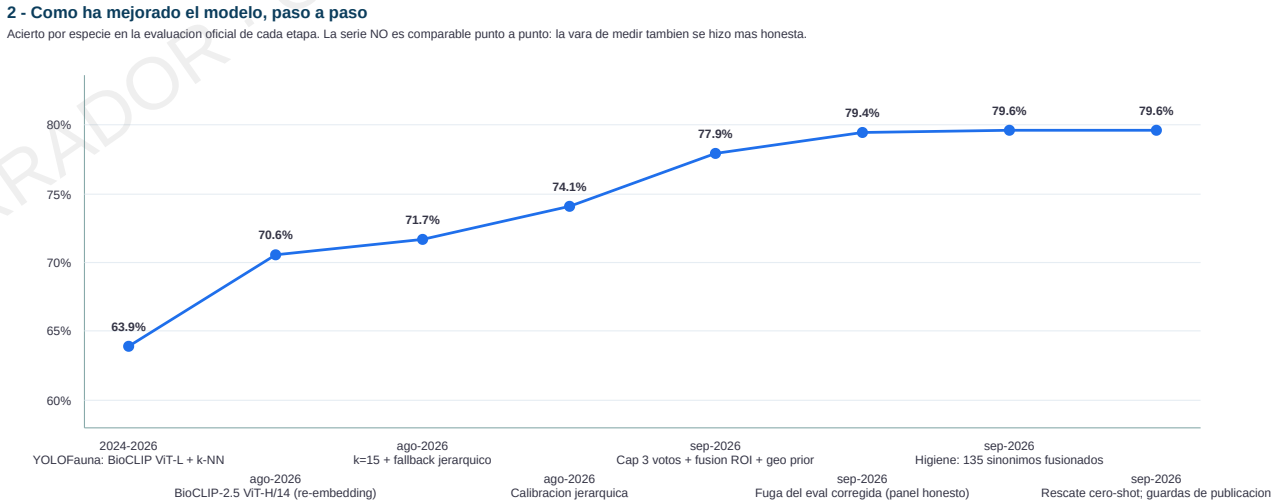
8. How BioFauna identifies a species

Pipeline: from a photograph to a calibrated, guarded answer

BioFauna is a **retrieval** classifier, not a trained end-to-end network:

1. **Frozen encoder.** Every gallery photograph is embedded with **BioCLIP-2.5 ViT-H/14** (frozen, no fine-tuning). The query image is embedded the same way, as the global vector **plus a 65 % region-of-interest fusion** (the subject-centred crop), L2-normalised.
2. **Approximate search.** A **FAISS** index holds all gallery vectors (currently **1.12 M vectors / 4,543 species**, row-aligned so that each vector maps to exactly one species).
3. **k-NN with a tempered aggregator.** The *k*=15 neighbours vote with weights `exp(similarity / T)`, `T=0.05`, **capped at 3 votes per species** so that a single over-represented species cannot swamp the vote.
4. **Context corrections.** A **geographic prior** (1,386 species, σ≈200 km) and **cryptic-pair** handling (Fisher directions / local subspace) correct the ranking when two species are visually almost identical.
5. **Hierarchical calibration.** A per-species logistic model converts the k-NN score into a **calibrated probability**, with genus- and family-level fallbacks; below the calibrated threshold the service **abstains** to genus/family instead of guessing.
6. **Zero-shot rescue (2026).** When the top-1 similarity is low (< 0.85), a **catalogue-wide zero-shot** comparison (2,985 text labels) is consulted; if it agrees with a *different* species at p ≥ 0.90, it wins. Measured effect: **+0.91 pp** on the field evaluation.
7. **Publication guards (AutoID).** A species is not published when the top-1/top-2 similarity **margin < 0.02**, when the observation falls outside the domain, or when a vocal disagreement exists; a **curator guard** automatically retracts our identifications when a curator corrects at class level or above.

9. How the model improved, step by step



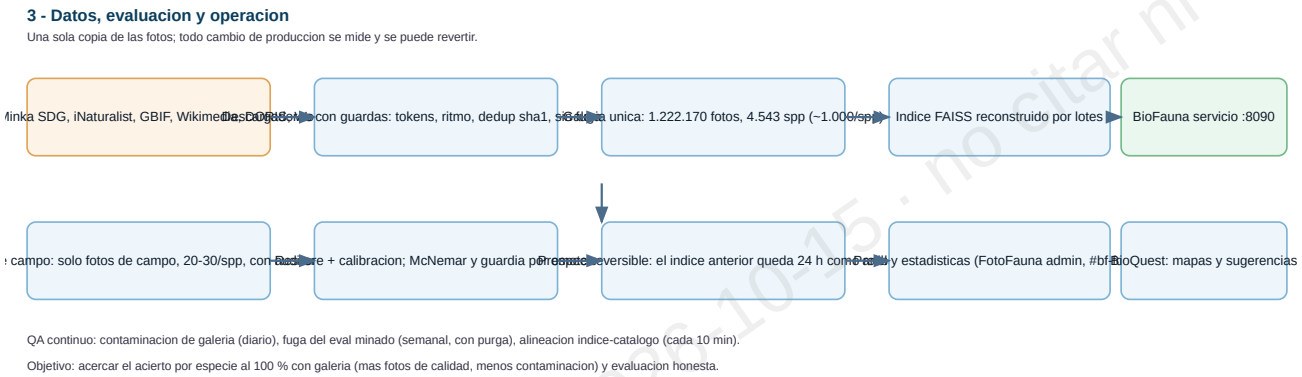
step	change	effect
YOLOFauna (2024–2026)	BioCLIP ViT-L + k-NN	63.9 %
Aug 2026	full re-embedding with **BioCLIP-2.5 ViT-H/14**	70.6 %
Aug 2026	**k=15** + hierarchical fallback + calibration hygiene	71.7 %
Aug 2026	**hierarchical calibration** (species/genus/family)	74.1 %

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step	change	effect
Sep 2026	**3-vote cap** per species, ROI fusion, geographic prior	77.9 %
Sep 2026	**evaluation leak fixed** → honest panel (near-100 % of the drop was leakage, not error)	79.4 %
Sep 2026	**gallery hygiene**: 135 duplicate synonym slugs merged (Minka authority), ~1,000 mislabelled photos relocated, 57 quarantined	79.6 %
Sep 2026	**zero-shot rescue** + publication guards	79.6 % *(+0.91 pp on the same rows; the headline moves with the evaluation)*

Two lessons that shaped the project: (a) **the evaluation is part of the model** — an unfixed leak in the evaluation made the system look better than it was; (b) **gallery quality beats model complexity** — synonyms, contamination and mislabelled observations cost more accuracy than any hyper-parameter we tuned (LoRA/QLoRA/ triplet/ArcFace/SupCon on this embedding space were all rejected by measurement).

10. Data, evaluation and operation



- **Sources** (public images): **Minka SDG** 377,476 · **iNaturalist** 779,411 · GBIF 39,570 · Wikimedia Commons 6,677 · DORIS/FFESSM 5,091 · SeaSlugForum 3,147 · WoRMS 1,320 · FishBase 1,072 → **≈1.22 M photographs**.
- **One copy of each photo**, never deleted; every relocation is done with a manifest and a rollback script.
- **Field evaluation**: field-only photographs (iNaturalist/Minka research grade), 20–30 per species, leak-audited before merging (a gallery copy in the evaluation would inflate the score).
- **Every production change is measured** (McNemar + per-species guard); the previous index stays as a **24-h rollback**.
- **Continuous QA**: gallery contamination (prototype-centroid metric, daily), mined-evaluation leak audit (weekly, with automatic purge), index ↔ catalogue alignment (every 10 min).

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Community taxonomic curators and expert reviewers; published checklists (GROC/OPK; Ballesteros 2007; Cervera et al. 2004; Salvador et al. 2022). Minka and iNaturalist observers. WoRMS. BioCLIP (Stevens et al.). The author also acknowledges the assistance of the main AI assistants available in 2026 — Claude (Anthropic), Grok (xAI), Gemini (Google), DeepSeek and others — which supported the engineering, evaluation and documentation of this system under the author's supervision.

Data availability

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Code, prototypes, catalog, calibrators, geo priors, cryptic pairs: this repository. Backbone: Hugging Face `imageomics/bioclip-2.5-vith14`. Images: obtain independently from Minka/iNaturalist/GBIF.

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Manuscript aimed at Biodiversity Data Journal / PeerJ. Artefacts and figures are in this repository.

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